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SUPERSEDING
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14 April 1983

MILITARY SPECIFICATION

BUSES, MOTOR: SUBURBAN AND INTERCITY, 41 TO 53 ADULT PASSENGER, 4x2, 6x2 AND 6x4, COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four-wheel, two-rear wheel drive; and six-wheel, two or four-rear-wheel drive diesel engine driven suburban and intercity type buses for long distance transportation with facilities for stowing passenger baggage. Buses include a rear or mid-mount diesel engine; single front and single or tandem rear axles. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in acquisition documents.

* 1.2 Classification. The bus shall be one of the types and passenger capacities shown in table I, as specified (see 6.2):

TABLE I. Vehicle classification.

Type	Wheels and wheels driving	No. of passengers *
I - Suburban	4x2	45
	6x2 or 6x4	53
II - Intercity	4x2	41
	6x2 or 6x4	49
* NOTE: Number of passengers is reduced by not more than three when a lavatory is required (see 3.5.11.1).		

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FSC-2310

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

*	A-A-393	- Extinguisher, Fire, Dry Chemical (Hand Portable).
	F-F-351	- Filters and Filter Elements, Fluid Pressure: Lubricating Oil, Bypass and Full Flow.
	L-P-383	- Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated.
*	ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).
	LLL-F-1238	- Floor, Covering, Linoleum.

MILITARY

	MIL-P-8013	- Plastic Materials, Polyester, Resin, Glass Fiber Base, Low Pressure Laminated.

STANDARDS
FEDERAL

*	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DEPARTMENT OF TRANSPORTATION (DoT)
Compliance with Interstate Motor Carrier Noise Emission Standards.
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Interstate Motor Carrier Noise Emission Standards.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* (Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government document which is current on the date of solicition.

SAE, INC.
SAE Standards and Recommended Practices

	J381	- Windshield Defrosting Systems Test Procedure - Trucks, Buses, and Multipurpose Vehicles.
	J382	- Windshield Defrosting System Performance Requirements - Trucks, Buses, and Multipurpose Vehicles.
*	J537	- Storage Batteries.
	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1,000 MHz).
	J588	- Turn Signal Lamps.

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	J589	- Turn Signal Switch.
	J688	- Truck Ability Prediction Procedure (DoD adopted).
*	J816	- Engine Test Code - Spark Ignitions and Diesel.
*	J1345	- Automotive Plastic Parts Specification.
	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

(Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.7, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All chassis items, body, and mounted equipment shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering office of the procuring activity prior to delivery of the items. The model furnished shall be not older than the manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting and marking. Treatment, painting, identification marking and data plates shall be in accordance with MIL-STD-1223. The

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exterior color and the identification markings shall be in accordance with MIL-STD-1223 for the military service identified by the procuring activity (see 6.2), except the vehicle roof to the drip molding line shall be painted the manufacturer's standard gloss white. When specified (see 6.2), concealed markings shall be furnished.

* 3.1.1.2 Heavy duty cooling system. A heavy duty cooling system shall be furnished that will maintain coolant at a temperature below the boiling point with the vehicle loaded to rated gross vehicle weight (GVW) and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 112 degrees Fahrenheit (°F) at sea level. Thermostatic control of engine coolant temperature shall be provided. Control of coolant temperature shall be accomplished by either thermostatically controlled radiator shutters, a thermostatically controlled variable pitch fan, or by thermostatically controlling the flow of all engine coolant through the radiator.

3.1.1.3 Coolant recovery system and level indicator. A radiator coolant recovery system or surge tank of adequate size to prevent loss of coolant under the conditions specified in 3.1.1.2 shall be furnished. A translucent reservoir, a sight gage, or other means for checking the coolant level shall be provided to enable the operator to check the coolant level while standing on the ground. In addition, a low coolant level and a high coolant temperature alarm buzzer and red indicator warning light shall be provided on the dash instrument panel.

3.1.1.4 Drain plugs. Drain plugs installed in the transmission and rear axle(s) shall be of the permanent magnet type.

3.1.1.5 Wood treatment. Wood flooring shall be treated in accordance with MIL-STD-1223 or with the manufacturer's standard commercial practice providing equivalent protection.

3.1.1.6 Towing device(s). Towing device(s) for towing the bus shall be provided on the front of the bus. The towing device(s) and the front of the bus shall withstand, without any sign of failure, turning the bus within its minimum turning radius and backing the bus on a level roadway.

* 3.1.1.7 Rustproofing. When specified (see 6.2), the chassis and underbody, including fenders, skirts and wheelwells, shall be rustproofed, if required on the type of body materials furnished, in accordance with the truck requirements of FED-STD-297.

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3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Interstate Motor Carrier Noise Emission Standards when tested in accordance with the Regulations of the Department of Transportation, Part 325.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis; the body with all attachments, accessories, and equipment; a full complement of fuel, lubricants and coolant; and an operator.

* 3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall be the combined curb weight and payload weight. The payload shall be the weight of the number of passengers specified based on 165 pounds per passenger and 165 pounds for the driver, plus baggage. The weight allowance for baggage shall be not less than 22.5 pounds per passenger.

* 3.2.6 Dimensions. Overall dimensions shall be as follows:

(a) Overall width shall not exceed 96 inches, exclusive of lights and mirrors;
(b) Overall height shall be not greater than 136 inches;
(c) Overall length shall be not greater than 40 feet 6 inches. Overall length shall include all front and rear protrusions except as otherwise specified herein;
* (d) Ground clearance shall be not less than 7 inches under axles and stabilizer rods and not less than 9-1/4 inches under all other parts.

3.2.7 Ratings. Vehicle ratings shall be manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity.

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* 3.2.8 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.3 Performance.

3.3.1 Speeds. High and low speed requirements shall be met with the bus loaded to specified GVW.

3.3.1.1 High speed gradeability. The bus shall ascend a 0.5 percent grade at 60 mph. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3). When air conditioning is furnished and powered by the main engine, the full load of the air conditioner shall be deducted in gradeability calculations.

3.3.1.2 Low speed. Low speed shall be calculated with engine operating at not less than 35 percent of recommended governed speed, and shall provide a vehicle speed not greater than 6.5 mph.

3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 68 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3)}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. Under all conditions of loading, the service brakes shall stop the bus, from a speed of 20 mph, within a distance of 35 feet from the point at which movement of the brake pedal begins. Deceleration and stopping distance shall be demonstrated on dry, hard, approximately level road surface that is free from loose material.

3.3.2.1 Downhill braking. The vehicle braking system shall include at least one of the following features:

(a) Brakes with a total of not less than 1,410 square inches of effective brake lining surface area for 4x2 buses and not less than 1,930 square inches of effective brake lining surface area for 6x2 and 6x4 buses;
(b) Automatic transmission with built-in hydraulic retarder;
(c) An engine type brake that brakes by engine compression through opening exhaust valves near piston top center and venting to the atmosphere. The engine brake shall include modulated control. The engine brake shall permit the bus loaded to GVW

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on a dry, hard road surface to negotiate a six percent downgrade without exceeding 27 mph, within the operating restrictions of the engine and transmission and without the use of service brakes.

3.4 Chassis.

* 3.4.1 Engine. The bus shall be furnished with a liquid-cooled, compression ignition, two-stroke or four-stroke cycle diesel engine with not less than six cylinders. The engine shall be the manufacturer's standard or optional engine for the commercial model vehicle which meets or exceeds the requirements of this specification. The engine shall be mounted at the rear or mid-mounted under the floor between the front and rear axles. An engine governor set and sealed to limit the engine speed to the maximum recommended operating speed shall be provided. An integral oil cooler may be furnished. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J816 or SAE J1349.

3.4.1.1 Engine compartment. The engine compartment ventilation and design shall allow operation without damage to engine, transmission and all components when the vehicle is operated in ambient temperatures of up to 112°F. Ready access to engine compartment shall be provided for servicing and routine maintenance of engine and engine components. Hinged engine compartment covers or doors shall be provided with a device to hold them open for servicing and maintenance.

* 3.4.1.2 Oil filter. A full-flow type oil filter shall be furnished, unless otherwise recommended by the engine manufacturer. The filter shall be connected directly to the engine lubrication system by flexible lines. The engine oil filtration system shall be approved by the engine manufacturer.

* 3.4.2 Starting system. The bus shall be furnished with a 12- or 24-volt starting system. Engine starting equipment shall include a glow plug or a pressurized ether system. The ether system shall include all equipment, including container mounting bracket, and any necessary piping. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type, controlled from the driver's compartment, with a reservoir of not less than 12 fluid ounces, and shall be inoperative with either the engine warm or the engine running.

3.4.2.1 Starter system controls. The starting motor shall be switch-operated by current fed (a) in parallel from a toggle switch in the engine compartment to both a start switch on the instrument panel and a start switch in the engine compartment and then to the operating

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coil of the relay, or (b) by current fed in series from a master run switch at the driver's instrument panel to a start switch in the engine compartment. (A safety switch shall be provided in the engine compartment so that the engine cannot be cranked from the driver's position when this switch is in the "OFF" position.) Means shall be provided to prohibit cranking of the engine except when the transmission is in the neutral position.

3.4.3 Alternator. When air conditioning is furnished and powered by the main engine, the alternator shall provide not less than 120 amperes at engine idle speed, and not less than 200 amperes at not more than 60 percent of the engine governed speed for a 12-volt system; or not less than 60 amperes at engine idle speed, and not less than 100 amperes at not more than 60 percent of the engine governed speed for a 24-volt system.

3.4.3.1 Independent 120-volt alternator. When an independent air conditioning system is provided with a separate 120-volt alternating current (ac) alternator driven by the auxiliary engine, the main alternator shall have a direct current (dc) output of not less than 90 amperes at engine idle, 125 amperes at 60 percent of engine governed speed, and 140 amperes rated capacity. When a 120-volt alternator is furnished, it shall have adequate power for operation of all air conditioning fans and blowers and shall have a rated output of not less than 1,200 watts at 120 volts ac, single-phase, 60 Hertz.

* 3.4.3.2 Alternator without air conditioning. When no air conditioning is furnished, the alternator shall provide not less than 90 amperes at engine idle and not less than 125 amperes at 60 percent of engine governed speed for a 12-volt system, or not less than 45 amperes at engine idle speed and not less than 63 amperes at 60 percent of engine governed speed for a 24-volt system.

3.4.4 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67 and shall provide all necessary piping, fittings, and accessories.

3.4.4.1 Air cleaner. An air cleaner(s) shall be furnished. The air cleaner(s) shall be of the dry type, with air restriction indicator to show when filter(s) should be replaced.

3.4.4.2 Fuel tanks. A fuel capacity of not less than 125 gallons shall be provided. When more than one tank is furnished, a balance line shall be provided. The filler shall be identified and arranged for safe and convenient filling from the exterior of the body. The access for the filler neck shall be located to prevent any accumulation of spilled fuel between the body panels or shall be sealed to prevent entry of spilled fuel to any point inside the exterior body panels.

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3.4.5 Fuel tank protection plate. If the fuel tank extends below the chassis frame, or body skirt, the tank shall be protected on the underside and at both ends by a steel plate of not less than 12 gage (0.1046 inch) thickness. The protection plate shall provide for drainage of any accumulated liquids and shall permit access to the fuel tank drain plug. A fuel tank protection plate is not required if a Federal Motor Carrier Safety tank is provided.

3.4.6 Transmission. Transmission input torque capacity shall be at least equal to the maximum torque delivered by the engine.

3.4.6.1 Automatic or semiautomatic transmission. Unless otherwise specified, the manufacturer's standard automatic or semiautomatic transmission recommended for the make and model coach shall be provided. The transmission shall provide required performance under all normal operating conditions and under all conditions specified herein. The transmission shall include a torque converter.

3.4.6.2 Manual transmission. When specified (see 6.2), the manufacturer's standard constant mesh manually shifted transmission shall be furnished. The transmission shall provide at least four forward speeds and one reverse speed.

3.4.6.2.1 Clutch. The clutch shall have a torque capacity exceeding the maximum delivered engine torque and shall be the largest capacity clutch offered for the engine and the vehicle model furnished. Pedal effort required to disengage the clutch shall not exceed 60 pounds. Spring or air assists, when employed, may be attached or incorporated within the clutch assembly.

3.4.7 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction. Propeller shaft guards shall be provided in accordance with Federal Motor Carrier Safety Regulation 393.89.

3.4.8 Chassis components. Chassis components shall be designed for bus construction. The body and the chassis shall be of the integral type. Body-on-chassis type construction, bolted or welded, is not acceptable.

* 3.4.9 Suspension. The vehicle shall be equipped with the manufacturer's standard full air or torsilastic type suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with vehicle loaded to GVW as specified in 3.2.5. The vehicle shall be provided with hydraulic

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double-acting shock absorbers: 2 on the front and 4 on the rear drive axle(s) for full air and not less than 2 on the rear drive axle(s) for torsilastic type suspensions. Shock absorbers shall also be provided at the rear non-drive axle, if manufacturer's recommended practice. The suspension design shall be such that the natural frequency of vertical vibration about any transverse axis shall be less than 1.5 cycles per second at rated capacity for the specified GVW. Outside rear wheels shall be designed and located so that there will be no interference with the use of tire chains. When full air suspension is furnished, the air system shall incorporate leveling valves with a time delay arranged to avoid constant air consumption due to road shocks. When full air suspension is furnished, the suspension shall be provided with a mechanism at each wheel to assure lifting of the wheel and axle when jacking the vehicle from the applicable jacking location (see 3.5.2). The suspension provided shall be approved by the bus body and chassis manufacturer.

3.4.9.1 Air springs. When full air suspension is furnished, the front axle shall be equipped with four air springs of 8 inches minimum diameter, or two air springs of 12-1/2 inches minimum diameter. Each rear drive axle shall be equipped with four air springs of 10 inches minimum diameter, and non-drive rear axle shall be provided with the manufacturer's standard air springs. All axles shall have none of the sprung weight carried on anything other than air springs. No other spring medium, such as leaf, coil or rubber springs, shall be used in conjunction with the air springs.

3.4.9.2 Radius rods and sway bars.

3.4.9.2.1 Radius rods (4x2 buses). Both front and rear axles on model 4x2 buses shall each be provided with not less than four radius rods to control lateral and longitudinal movement. Radius rods shall transmit driving and braking forces to the vehicle chassis and the body. Each end of the radius rods shall be equipped with rubber bushings that do not require periodic lubrication.

* 3.4.9.2.2 Radius rods (6x2 and 6x4 buses). When full air suspension is furnished, the front axle of model 6x2 and 6x4 buses shall be provided with not less than four radius rods to control lateral and longitudinal movement. Radius rods shall transmit driving and braking forces to vehicle chassis and body. The rear axle tandem assembly shall be the manufacturer's standard. The tandem assembly shall be provided with not less than eight radius rods: four to control longitudinal movement and four to control lateral movement; or the drive axle shall have four radius rods to control lateral and longitudinal movement. The non-drive axle shall be of an independent or swing arm suspension design

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incorporating adequate means to control lateral and longitudinal loads such as upper and lower "A" frames and an adjustable toe-in link. Each end of the radius rods shall be equipped with rubber bushings that do not require periodic lubrication.

* 3.4.9.2.3 Sway bars. When full air suspension is furnished, sway bars of adequate size and strength shall be incorporated in both front and rear suspension systems or transverse front and wishbone type radius rods at the rear may be provided for lateral stability.

* 3.4.9.3 Torsilastic type suspension. When torsilastic type suspension is furnished, it shall have been the manufacturer's standard suspension for at least the past ten years. The suspension shall be warranted for not less than 3 years or 300,000 miles. The suspension shall be torsilastic type suspension having rubber springs on all axles. The rubber compound shall be compounded to provide torsional springing action. The torsilastic unit shall consist of an outer steel tube and an inner splined tube with the rubber compound bonded to the outside wall of the spline tube and to the inside wall of the outer tube. On the front and drive axles, a smaller spline tube shall be attached to the chassis frame with a larger outside tube attached to the traveling portion of the suspension. On bogie trailing axles, the larger outside tube shall be attached to the chassis frame with a smaller spline tube traveling with the suspension. The front and rear suspension systems shall provide full vehicle stability under all operating conditions.

* 3.4.10 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, when the vehicle is loaded to GVW as specified in 3.2.5. Model 6x2 and 6x4 vehicles shall be provided with two rear axles, one or both being drive axles. Single reduction drive axle(s) shall be furnished. When specified (see 6.2), oil seals shall be provided on all the axles. The wheel bearings and axle spindles shall be oil lubricated. The hubcaps shall have a window for visual determination of the oil level.

3.4.10.1 Traction control. When specified (see 6.2), 4x2 or 6x2 vehicles shall be furnished with a traction control on the drive axle. The traction control shall actuate automatically to insure that power is transmitted to the wheels having traction when the opposite wheel loses traction. When a 6x4 vehicle is furnished, the dual drive axles shall incorporate a means of locking out all differential action, providing full traction on all rear wheels. The locking and unlocking device shall be operated by a switch within easy reach of the driver in the normal seated position. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

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3.4.11 Wheels, rims and tires. The vehicle shall be equipped with single wheels on the front axle and with dual wheels on the rear drive axle(s). Tire size, tread design and load range (ply rating) shall be the same for all tires on the front axle and rear drive axle(s). When a non-drive rear axle is furnished, dual or single wheels may be furnished. Tire and rim size may be smaller than those furnished for the drive axle, but shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with the vehicle loaded to the GVW specified in 3.2.5. Rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.11.1 Tires. Tires shall be tubeless steel belted radial type with highway tread. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded to GVW as specified in 3.2.5. When a non-drive axle is furnished and the tires supplied on the non-drive axle are smaller than those on the drive axle, the tires supplied on the non-drive axle are not required to be steel belted radial design. Tires shall conform to Tire and Rim Association recommendations, or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

3.4.11.2 Balancing. Each tire shall be balanced within practicable limits. Wheels, hubs, and brake drums shall be effectively in balance. Balancing shall be adequate to preclude wheel shimmy at all vehicle speeds.

3.4.11.3 Spare wheel. When specified (see 6.2), a spare wheel or rim for the front and for the rear drive axle(s) shall be mounted on a carrier and installed in a readily accessible location. The carrier design shall enable removal or mounting of the spare wheel assembly using only the tools specified in 3.4.18. Threaded fasteners, when used to secure the spare wheel or rim or the spare tire assembly in the carrier, shall be constructed of, or plated with, corrosion-resistant material. The carrier and spare shall not be located in the baggage compartments. The spare wheel may be stowed in a suitable separate compartment.

* 3.4.11.4 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design, and load range as the tires in use on the front axle and the rear drive axle(s).

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3.4.12 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42(b) and 393.44 through 393.52.

* 3.4.12.1 Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air- or water-cooled, and having a capacity of not less than 13 cubic feet per minute (cfm)
(b) Air storage reservoir(s) with a total capacity of not less than 3,800 cubic inches for 4x2 vehicles and 4,200 cubic inches for 6x2 and 6x4 vehicles. Each tank shall be equipped with drain, and with safety and check valves between the compressor and the last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low air-pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
* (i) Ping tank or discharge muffler, if necessary for noise control
(j) Automatic slack adjusters.

NOTE: When an air dryer is furnished between the air compressor and the tank, (h) the automatic moisture ejector shall be omitted.

3.4.13 Electrical system. A 12-volt, 24-volt, or combination 12-volt/24-volt electrical system shall be furnished. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.13.1 Lighting. All vehicle lights, reflectors and wiring shall be as specified herein and shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20, and 393.23 through 393.26. Lights and reflectors shall not be mounted on rub rails (unless recessed and fully protected) or the vehicle bumpers. An illuminated rear license plate holder shall be provided. Engine compartment illumination consisting of not less than two light fixtures connected to one switch shall be provided. Fixtures shall be placed so as to provide uniform compartment illumination. Back-up lights shall be provided.

3.4.13.2 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. The vehicle shall be provided with single-faced front signal units and single-faced rear signal units

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installed in accordance with SAE J588. Turn signal operating units shall be of the self-cancelling type with visible and audible flash indicator. An additional set of armored type amber turn signals, mounted on each side of the bus above the front wheelwells, shall be furnished.

* 3.4.13.3 Passenger signal buzzer. Unless otherwise specified (see 6.2), signal buzzer equipment or an electric chime shall be furnished, operated by pull cords or continuous type switch on both sides of the bus, or by one rocker-switch type button call for each set of two seats, and provided with a driver's cutout switch.

* 3.4.13.4 Interior lighting. The manufacturer's standard general lighting fixtures arranged to provide general illumination from the ceiling area along both sides of the bus shall be provided. The general lighting shall consist of not less than seven bulbs of not less than six candlepower each, arranged in accordance with the manufacturer's standard for the model coach furnished. Equivalent fluorescent light tubes are acceptable for general lighting. Type II buses, and when specified (see 6.2) type I buses, shall have individually controlled reading lights of not less than 10 candlepower over each seat except the center seat of the 5-passenger rear cross seat, and over the front row of the passenger seats if such seats are on a lower level. A domelight, actuated by opening the service door or actuated by the driver, shall be provided for illumination of the driver area. Door-operated step lamp(s) shall be installed to provide illumination of the stepwell and of the ground adjacent to the step.

3.4.13.5 Spotlight. When specified (see 6.2), a spotlight shall be furnished on the left front corner of the bus, positioned to be operated from the driver's seat and to illuminate the area in front of the door.

* 3.4.14 Batteries. Batteries shall be of 12- or 24-volt potential. Total capacity of all batteries shall provide not less than 400 ampere-hours for 12 volts and 200 ampere-hours for 24 volts at a 20-hour rate. Battery ratings shall be in accordance with SAE J537. Batteries shall be accessible for inspection without detaching the batteries from the vehicle. The batteries may be mounted in the engine compartment, on trays, or on the battery compartment floor. The battery tray or battery compartment floor shall be acid-resistant stainless steel or shall be coated or covered with acid-resistant material. When batteries are mounted in a special battery compartment, the compartment shall be large enough to inspect and service the batteries without requiring their removal from the coach. A battery access door shall be provided on the side of the body. Access to the batteries through the baggage

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compartment door is acceptable. If battery trays are used for inspecting the batteries, they shall swing out or slide out on rollers, if necessary, to support the batteries. The batteries shall be located away from or protected from engine/transmission heat that will have an adverse effect on battery life or operation.

* 3.4.14.1 Battery cables. Battery cables shall be routed and secured in such a manner as to prevent chafing the cables at any point. Insulated studs or rubber grommets shall be used where cables pass through the frame metal members and the bulkheads. Cables shall be so mounted, fastened and protected that they will not become loose during service.

3.4.15 Steering. Power steering shall be furnished. Power steering tubing, fittings, and mechanical connections shall be located to provide easy access for inspection and servicing.

* 3.4.16 Windshield wipers and washers. The vehicle shall be equipped with dual, air operated windshield wipers of the parallel type. The wiper system furnished shall deliver not less than 32 pound-feet of torque and sweep the maximum practicable area of the windshield surface. Windshield wiper arm tip force shall be not less than 36 ounces. Air operated windshield wiper motors shall be "Sprague Air Push II" or equivalent, with 5/8-inch diameter drive. Wipers shall conform to Federal Motor Carrier Safety Regulation 393.78. Windshield washers shall be provided. Washers shall be electric or air operated with a control on the instrument panel.

3.4.17 Bumpers. Front and rear bumpers shall be provided. The bumpers shall be of anodized aluminum section, stainless steel, or chrome-plated steel. The rear bumper shall conform to Federal Motor Carrier Safety Regulation 393.86 and shall be as specified in 3.5.11.9.

3.4.18 Tools. Unless otherwise specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly (see 3.4.11.4), and shall include at least a hydraulic jack, jack handle, and wheelnut wrench. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly. When jack height exceeds the height under the jacking location with the tire flat, run-up blocks shall be furnished. Storage provisions for the tool equipment and tire chains shall be furnished. The stowage provisions shall be accessibly located and weatherproof.

3.4.19 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls

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shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's seat. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be provided with indirect illumination. The instrument panel shall be designed to preclude windshield reflection of instruments.

* 3.4.20 Accessories and equipment. The vehicle shall be complete with at least all the controls, indicators, and gages furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Speedometer
(e) Recording odometer or hubodometer
(f) Oil pressure gage with visible and audible warning for low oil pressure
(g) Engine coolant temperature gage with visible and audible warning for overheated engine
(h) Indicator light to show headlight high-beam operating
(i) Indicator light to show operation of stoplights
(j) Turn signal pilot indicator
(k) Transmission high temperature gage, when an automatic transmission with a retarder is furnished
(l) Low transmission oil pressure warning light and buzzer when a manual transmission with a pressurized oil system is furnished
(m) Tachometer.

* 3.4.20.1 Horn. An electric horn shall be provided.

* 3.4.21 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

* 3.4.22 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. The exhaust tailpipe shall extend to the rear bumper and shall be so located as to minimize the possibility of exhaust fumes entering the body.

3.5 Body. The bus shall provide seating for the specified number of adult passengers. The body shall have full-skirted sides, shall present a generally symmetrical appearance, shall exclude engine fumes

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and exhaust gases from the bus interior, and shall provide safe and comfortable transportation for the passengers and the driver.

* 3.5.1 Inside body dimensions. At seat level, body inside width shall be not less than 89-1/2 inches. Between the first and last vertical roof bows through the center of the aisle, inside height shall be not less than 75 inches.

* 3.5.2 Construction. The body and the chassis shall be of integral type construction. The body and the roof framing shall be constructed of metal sections with insulated aluminum skin. All parts, where applicable, shall be fastened together in a manner which will preclude loosening of bolts, screws, rivets, and cracking of welded joints when the vehicle is operated under intended service conditions. Jacking locations shall be provided at each wheel. Plates shall be of sufficient capacity to withstand the jacking force required to lift the portion of the loaded vehicle affected without damage to any vehicle component. Self-tapping screws shall not be used where subjected to stress. Wood shall not be used for structural framing. Outside body panels and other sheet metal shall be of aluminum alloy or stainless steel of not less than the gages, as applicable, regularly furnished by the manufacturer in a bus of the model offered. The body, roof, and panel joints shall be watertight and leakproof with leading panels overlapping the edge of trailing panels. The pitch of fasteners (bolts, screws, and rivets) at all major exterior sheet metal panel edges shall not exceed 24 times the thickness of the thickest material used in the joint.

3.5.3 Floor. The floor frame and floor plates, with entrance steps and wheelhousings, shall be of substantial construction and shall conform to Federal Motor Carrier Safety Regulation 393.84. Wood flooring of not less than 1/2-inch, 5-ply marine grade or Douglas fir improved select sheeting, exterior grade plywood shall be furnished. Wood parts shall be treated as specified in 3.1.1.6. Steel or extruded aluminum channels, ranging in size from not less than 1-1/8 to 2 inches in depth and spaced to provide support for the loads imposed, shall be furnished under the floor.

* 3.5.3.1 Stepwell. The stepwell shall be of the enclosed type with two, three, or four steps, and constructed of patterned or lined anodized aluminum, stainless steel or fiberglass. The height of the bottom step shall not exceed 17 inches with the bus unloaded. Riser heights shall provide equal step height within 1/2 inch. Step covering shall be not less than 5/16-inch thick ribbed rubber with white molding, or ribbed stainless steel nosings.

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3.5.3.2 Wheelhousings. Wheelhousings shall include splash shields, and shall provide ample clearance for use of tire chains on the outside duals. Molded rubber or metal splash trim shall be installed around the inside or outside lip of the outside edge of both the front and rear wheelhousings.

3.5.3.3 Floor coverings. The under seat floor area and the wheelhousings shall be covered with not less than 3/32-inch thick rubber or synthetic rubber blend material, or linoleum conforming to type I, class 1, grade B of LLL-F-1238. The entrance area and steps shall be covered with not less than 3/16-inch thick rubber or synthetic rubber blend ribbed material. Coverings shall be sealed at the joints and bonded to the undersurface using a waterproof adhesive. The floor in the driver's area shall be covered with a mat or with the same materials as used in the under seat or aisle area. The aisle shall be covered with not less than 3/32-inch thick rubber or synthetic rubber blend material.

3.5.4 Sides and roof. The sides and the roof shall have sheet metal outer panels, lined on the inside and integrated with the body structure. The front and rear roof cap sections may be constructed of unstressed molded fiberglass sections in lieu of sheet metal. The vehicle front cowl section and rear end section shall be metal outer panels or fiberglass reinforced thermosetting plastic (see 3.5.4.1.1).

* 3.5.4.1 Exterior. The sides shall be equipped with rub rails. When integral with the side panels, the rub rails shall run at least between the front and rear wheelwells. When rub rails are not integral with the panels, rub rails shall run the full length of each side of the coach. The side panels shall be a combination of fluted stainless steel or fluted anodized aluminum panels and smooth painted aluminum panels. Front and rear panels shall be of the same construction except reinforced plastic may also be used.

3.5.4.1.1 Fiberglass reinforced plastic front cowl and rear end section. When fiberglass front and rear sections are furnished, the sections shall be not less than 0.210 inch in thickness. When fiberglass front and rear roof cap sections are furnished, the sections shall be not less than 0.120 inch in thickness. The material shall conform to type I of L-P-383 and shall have the minimum mechanical properties for "Mat" fabric listed in table I of MIL-P-8013. A polyester resin and fiberglass depositor may be employed to apply the resin and fiberglass. The plastic sections shall be composed of not less than 40 percent (by weight) glass fibers, and additives to the resin mix shall provide for fire-retardant and weathering performance properties. Color additives need not be added to the resin mix.

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* 3.5.4.2 Interior. The body interior, including the sides, roof, and front and rear ends, shall be fully lined with metal or other substantial, puncture-resisting, hard surface lining covered with ABS (acrylonitrile/butadiene/styrene) plastic material, in accordance with SAE J1345. The material shall be washable and shall blend with the interior finish of the bus. Interior metal trim shall be anodized aluminum or stainless steel. The interior color of the vehicle lining and upholstery shall be compatible with the exterior vehicle color furnished.

3.5.4.3 Insulation. Fiberglass mats or blankets shall be securely installed in roof, side, and end walls.

3.5.4.4 Destination sign compartment. Unless otherwise specified (see 6.2), a destination sign compartment shall be installed in the front. The compartment shall provide not less than 130 square inches of glass enclosed frontal area, of a height to permit reading of 4-inch letters on the sign roller, and shall be readily accessible through a latched door from inside the bus. The destination sign shall be of the 15-listing (furnished blank), 4-inch letter, roller type, operable from the interior of the bus. The compartment shall be illuminated to permit reading of the destination sign at night. The access door shall preclude entry of exterior air or moisture to the interior of the bus through the destination sign compartment. When the destination sign frame and the surrounding roof cap is molded of fiberglass as specified in 3.5.4.1.1, the minimum wall thickness shall be not less than 0.110 inch.

* 3.5.5 Service door. The service door shall be on the right-hand side located toward the front of the bus. The door shall be provided with compression or overlapping seals to exclude dust, water, and cold air. The service door shall be sedan type or pantograph type with a manual or air operated control conveniently located for driver operation and shall have a key-operated locking device or shall be provided with means for locking with a padlock. The service door shall open outward. The service door shall be provided with safety glass panel(s). Minimum dimensions of clear opening shall be 24 inches in width at center, 16-1/2 inches at the top, and 74 inches in height. Padding shall be installed above the door to prevent head injury. The bus shall be equipped with an exterior release control to open the door from the exterior of the bus.

3.5.6 Windshield. A safety glass windshield affording maximum practicable driver vision shall be installed at a slope sufficient to minimize external glare or reflections from inside the body. Glass shall be tinted.

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* 3.5.7 Windows. Each side window in the passenger area shall be not less than 51-1/2 inches long, measured along the center of the glass area, and not less than 19 inches high. Sash material shall be anodized aluminum or stainless steel, and shall conform to the best current commercial practice. If air conditioning is not furnished, passenger side windows shall be of the horizontal sliding type with fingertip control, permitting ready opening and adjustment to various desired openings. Windows shall be rattleproof, rainproof and windproof. The driver's window shall be of the horizontal sliding type or shall include a horizontal sliding section in the lower rear corner to allow for signaling and for ventilation. Safety glass shall be used in all windows. All windows shall be fitted with locking devices. At least every other window on each side of the bus shall be of the push-out type and hinged at the top, for use as emergency exits. In addition, one roof hatch shall be installed at each end of the coach, for use as emergency escape hatches and for ventilation purposes.

* 3.5.7.1 Tinted windows. Unless air conditioning is not required, tinted, heat absorbing, safety window glass shall be furnished. Entrance door windows, driver's left window(s) and windshield shall be in accordance with the manufacturer's standard for tinted glass. Tinting of lamination is not acceptable except for dark bands not more than 12 inches wide, that do not restrict the driver's vision, at the top of the driver's and entrance windows, if manufacturer's standard. Side windows in passenger area shall be tinted to double density or shall be of the graduated tinted type or shall be tinted to allow not more than 33 percent light transmission. Rear window(s) shall be tinted to double density, allowing not more than 33 percent light transmission, or a solid insulated nontransparent panel shall be furnished. For passenger area side and rear windows, tinted glass with tinted lamination may be used.

3.5.8 Seating.

* 3.5.8.1 Type I (suburban). Suburban type seating shall be provided to accommodate the specified number of adult passengers and the driver. All seats shall be of the forward-facing type on not less than 30-1/2 inch centers and shall be mounted on each side of a center aisle. The bus shall have at least ten 2-passenger seats on each side for model 4x2 vehicles and twelve 2-passenger seats on each side for model 6x2 and 6x4 vehicles. Unless a lavatory is specified (see 3.5.11.1), the manufacturer's standard five-passenger full width rear cross seat shall be provided. The seat back of the rear cross seat shall be installed at the same angle of inclination as the regular two-passenger seats. The front seat cushions shall be spaced at least 10 inches from the modesty panel and 9 inches from the back of the driver's seat, with the driver's

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seat adjusted approximately 17 inches from the seat back to the steering wheel and the driver's seat back in the medium tilt position. Seats located at or near wheel housings shall be installed to minimize interference with passenger comfort. All seats shall conform to Federal Motor Carrier Safety Regulation 393.91 and shall be secured to prevent shifting under all operating conditions. The 2-passenger seats shall be American Seating Company Model 6592 or equal.

3.5.8.1.1 Passenger seat frames (type I). Frames of passenger seats shall be constructed of formed metal tubing, and shall include armrests and footrests (type I, style A). The seats shall be the manufacturer's standard 36-inch width. The seat cushions shall have a free depth of not less than 17 inches from a vertical plane at the front edge to a vertical plane at the most forward part of the seat back cushion and shall be individually constructed with full depth polyether foam or polyurethane foam. Seat backs shall be of full depth polyether foam or polyurethane foam with a free height of not less than 26 inches from the seat cushion. The top, the upper corners, and not less than the upper 6 inches of the rear surface of each two-passenger seat back shall be constructed of or covered with force distributing material. The material shall minimize force and spread the area of contact upon impact by an occupant in the next seat to the rear. The rear surface of the seat back below the force distributing material shall be covered with stainless steel or scuff- and impact-resistant plastic. Closure type ash receptacles shall be permanently installed in locations convenient to passengers in the two front seats, and on the back of each two-passenger seat.

* 3.5.8.2 Type II (intercity). Type II buses shall be provided with deluxe intercity type seating to accommodate the number of passengers specified. All seats shall be of the forward-facing type on not less than 32-1/2 inch centers. Two-passenger reclining type seats shall be provided on each side with the front seat cushion spaced at least 14 inches from the stairwell modesty panel and 14 inches from the driver's modesty panel; except the spacing shall be not less than 12.5 inches from the driver's modesty panel on 49-passenger buses. Unless a lavatory is specified (see 3.5.11.1), the manufacturer's standard five-passenger, full width rear cross seat shall be provided. Seats located at or near wheelhousings shall be installed to minimize interference with passenger comfort. All seats shall be secured to prevent shifting under all operating conditions. All two-passenger seats may be mounted on a recessed track, to permit varied arrangement of seats. When seats are mounted on a recessed track, a plastic filler strip shall be provided to close the voids between seat mounting points.

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* 3.5.8.2.1 Passenger seat frames (type II). Frames of the passenger seats shall be constructed of formed metal tubing, wall mounted, with legs in center or on the aisle side, or shall be "T" pedestal mounted. Each passenger seat shall include armrests and a two-position adjustable footrest. The 2-passenger seats shall be American Seating Company Model 6681, National Seating Company Model 3270, or equal. The seats shall be the manufacturer's standard nominal 38-inch width. Seat cushions shall be not less than 17 inches in depth from a vertical plane at the front edge to a vertical plane at the most forward part of the seat back cushion. Seat backs and seat cushions shall be constructed with full depth polyester foam or polyurethane foam, and shall include a headrest. All 2-passenger seats shall be of the individual type. The top, the upper corners, and not less than the upper 6 inches of the rear surface of each 2-passenger seat back shall be constructed of or covered with force distributing material. The material shall minimize force and spread the area of contact upon impact by an occupant in the next seat to the rear. The rear surface of the seat backs below the force distributing material shall be covered with scuff- and impact-resistant plastic. Closure type ash receptacles shall be provided for each 2-passenger seat, except for the front aisle seats and the center of the rear cross seat.

* 3.5.8.3 Seat covering. Seat covering material (types I and II) for the passenger and driver's seats shall be combination woven fabric and vinyl plastic. Seat woven fabric shall be not less than 42 ounce weight per linear yard and vinyl plastic shall be not less than 40 ounce weight per linear yard. The top of the seat cushion and the front of the seat back shall be covered with woven materials of identical pattern and design, except the force distributing material shall be covered by vinyl. Side boxing of seat cushions and seat shall be covered with vinyl material of a color coordinated with the fabric material utilized. The headrest and force distributing material shall be covered with vinyl plastic. Seat and back covering shall be readily removable. Seat covers shall be readily removable without taking armrests apart and shall be able to withstand dry cleaning.

* 3.5.8.4 Driver's seat. The driver's seat shall be heavy duty bus type, pedestal mounted and adjustable fore-and-aft and for height without the use of tools. Seat cushions, padded with foam rubber, polyether foam or polyurethane foam, shall be provided with tilting adjustments for cushion and back to provide maximum driver comfort. All these adjustments shall be capable of being made manually by the driver while he is seated and shall be safeguarded against pinching of fingers. The adjustment mechanism shall be sufficiently rugged to hold adjustment in service, shall be properly shielded from dirt, shall be located so as not to be a hazard during operations and shall preclude accidental raising or lowering. A lap type seat belt shall be installed for the driver's seat.

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3.5.9 Stanchions, modesty panels and grab rails. Stanchions and grab rails of stainless steel, stainless clad steel, or plastic coated steel tubing, at least 1-1/4 inches in diameter, shall be installed. The grab rails in the service door area shall be not less than 1 inch in diameter. A slide or roll type opaque curtain or cross rail routed to the rear of the driver from the closure panel to the right of the driver to the body side shall be installed to provide protection for the driver from passenger interference. A complete modesty panel behind the driver or a closure panel to the right of the driver shall be provided to prevent the right front aisle passenger from accidentally disengaging the gear shift. A stanchion or grab rail with modesty panel shall be provided at the rear of the service door entrance and a stanchion or grab rail shall be provided at the front of the entrance, arranged to assist passengers entering or leaving the bus.

3.5.10 Heating, defrosting, ventilating and cooling. Heating, defrosting, ventilating and cooling systems shall conform to Federal Motor Carrier Safety Regulations 393.77 and 393.79.

3.5.10.1 Heater. The main system shall provide filtered, fresh, and recirculated airflow through a common duct system. The duct system shall disperse air through louvered vents below the sash of all side windows or through fin tube ducting installed at the junction of the floor and both side walls, and extending the entire length of the bus passenger area. A dash system shall furnish heat in the area of the driver, windshield, and door. Engine coolant hot water type heater(s) shall be provided. The capacity of the heating system shall maintain the inside bus temperature at a minimum of plus 70°F, midway between floor and ceiling, in ambient outside air temperature of 0°F, with no passengers inside. The temperature gradient between the floor and the ceiling shall not exceed 10°F. Blower capacities shall provide a ventilating airflow of at least 1200 cfm. The system shall be provided with thermostatic controls.

* 3.5.10.2 Defroster. Defroster outlets with suitable connections and independent blower(s) shall be furnished. The independent blower(s) shall have a rating of not less than 250 cfm, and shall be installed to provide a flow of heated air to the full windshield at not less than 23,000 British thermal units (Btu) per hour. An adjustable auxiliary fan shall be provided on top of the dash with a guard. The windshield defrosting system shall conform to SAE J381 and SAE J382.

3.5.10.3 Ventilator. A manual, adjustable, flow type ventilator opening shall be provided at the front of the bus. The outside openings for heaters and ventilators shall be so located as to minimize entry of fumes, road dust, rain, or road wash and shall be covered in such a way as to prevent clogging in bad weather.

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* 3.5.10.4 Air conditioning. Unless otherwise specified (see 6.2), the vehicle shall be equipped with the manufacturer's standard air conditioning system. The manufacturer shall furnish to the engineering department of the procuring activity the rated capacity at the specified conditions of the entire unit, either by air measurement or by the flowmeter method. The air conditioner shall have a cooling capacity of not less than 72,000 Btu per hour and shall be capable of maintaining not more than an 80°F dry bulb and a 68°F wet bulb temperature and an effective temperature of 74°F inside the bus when the ambient temperature is 95°F dry bulb. The humidity shall be not less than 30 percent and not more than 60 percent. Inside temperature shall be measured midway between floor and ceiling at the front, middle, and rear seats with no passengers in the bus and not more than a 6°F wet bulb temperature gradient within the bus. The cooling performance specified herein shall be met when the bus is traveling at 65 mph and when standing still with the main engine at a fast idle. The air conditioning system shall be integrated with the heating system to the maximum extent possible, and shall automatically proportion air in the ratio of not less than 8 percent fresh air, with the rest of made up of recirculated air. All air shall pass through the filter system. Air conditioner controls shall be panel mounted within easy reach of the seated driver.

3.5.10.4.1 Independent system. When an independent air conditioning system is furnished, it shall be powered by a diesel engine of not less than 35 horsepower rating. Engine fuel shall be from the main engine fuel tank. All air conditioning components except the evaporator shall be located in a compartment under the floor. Air conditioning engine starter, safety control switches to cut out driver's controls, compartment lamp switch, and junction panel shall be installed in the auxiliary engine compartment, in addition to controls provided at the driver's seat. The air conditioning shall be designed for bus operation and shall be complete with all necessary controls for automatic operation of the unit after the power unit has been started and the compressor engaged. The air conditioning system may include a 110-volt ac alternator, driven by the auxiliary engine, to power air conditioning blowers.

3.5.11 Body equipment. Body equipment shall be complete with all accessories listed by the manufacturer as standard equipment and in addition shall include items specified in 3.5.11.1 through 3.5.11.9.

* 3.5.11.1 Lavatory. When specified (see 6.2), the bus shall be furnished with the manufacturer's standard lavatory facility including at least a flush toilet, a washstand and a waste holding tank. A light and power vent with switch(es) shall be furnished. The lavatory

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facility shall be installed at the curbside rear of the bus and the five-passenger rear seat may be replaced with the manufacturer's standard two- or three-passenger seat. Total passenger capacity may be reduced by not more than three passengers.

3.5.11.2 Highway warning kit. Three emergency reflective triangles conforming to Federal Motor Carrier Safety Regulations 393.95(f)(2)(i) shall be furnished and properly stowed.

3.5.11.3 Stowage compartment(s). Provision shall be made for weatherproof stowage of tools, emergency reflective triangles and tire chains under lock and key or concealed latch, or provided with means for locking with a padlock.

3.5.11.4 Keys. Duplicate keys shall be furnished for all key operated locks or padlocks. Keys shall be identified, and secured to vehicle steering wheel or otherwise stowed in an approved manner for shipment. All locks on the same vehicle shall be of the same key code, except the ignition key.

3.5.11.5 Standee line. Areas prohibited to standees shall be marked in accordance with Federal Motor Carrier Safety Regulation 393.90.

3.5.11.6 Rearview mirrors. One nonglare type mirror having not less than 49 square inches of reflective area or a convex mirror not less than 35 square inches in area shall be mounted inside the bus. Two rearview mirrors shall be mounted outside at the front of the bus. The reflecting surface of each outside mirror shall be not less than 8 inches by 8 inches in size. Outside mirrors shall be mounted one each on the left and right sides of the bus and shall have retractable arms. Mirrors shall be friction mounted to prevent damage when struck.

3.5.11.7 Sunvisor. An adjustable sunvisor shall be furnished.

* 3.5.11.8 Emergency equipment. Emergency equipment shall include first aid kit provisions, a small fire axe, and a fire extinguisher conforming to Federal Motor Carrier Safety Regulation 393.95(a). Emergency equipment shall be installed near the front of the coach convenient to the driver. The fire extinguisher shall be a minimum 2-1/2 pound, dry chemical type; shall have an Underwriter's Laboratory rating of 5B:C or more; shall include a metal safety pin and wire; and shall conform to type I or type III, class 2, size 2-1/2 of A-A-393. The fire extinguisher shall permit visual determination as to whether it is fully charged. Not less than a 9-unit type first aid kit shall be furnished or when specified (see 6.2), a bracket shall be provided for

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future installation of a 12-unit type first aid kit. The 12-unit type first aid kit size is 9-7/16 inches long, 5-7/16 inches wide, and 2-7/8 inches thick. The 12-unit type kit is not required to be furnished with the vehicle.

3.5.11.9 Rear bumper. The rear bumper shall have anti-ride and anti-hitch features.

3.5.11.10 Interior public address system. An interior public address (PA) system shall be furnished. The PA system shall include a switched microphone, an on/off control and a volume control conveniently located on the driver's console. The speakers shall be located so that the PA system is audible throughout the bus while the bus is in motion.

* 3.5.12 Package racks. Type II buses and unless otherwise specified (see 6.2) type I buses shall have panel type package racks on each side of the bus over the two-passenger seats on the main floor level. Package racks shall be not less than 20 inches wide and shall include the manufacturer's standard structural members.

* 3.5.12.1 Rack panel material. Package racks shall be of the panel type, of the manufacturer's standard material compatible with the decor of the coach. When specified (see 6.2), enclosed package racks with doors shall be furnished.

3.5.13 Baggage compartments. When a mid-mounted engine is furnished, the baggage compartment(s) shall provide not less than 145 cubic feet of usable space if no air conditioning is furnished, or 125 cubic feet if air conditioning is provided. When a rear engine is furnished, the baggage compartment(s) shall provide not less than 212 cubic feet of usable space. All baggage compartments shall be adequately illuminated, shall incorporate a switch, and shall be provided with means for secure locking.

3.6. Servicing and adjusting. Prior to acceptance of the vehicle by the Government, the contractor shall service and adjust the vehicle for operational use including at least the following: alinement of lights; adjustments of the engine and brake systems; filling and charging of the battery; alinement of the front wheels; inflation of all tires; complete lubrication of body, chassis, engine, and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of the windshield washer reservoir with water and appropriate additives.

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3.7 Workmanship. Defective components, or parts and assemblies which have been repaired or modified to overcome deficiencies, shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

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4.3.1 Heating and air conditioning. The vehicle shall be tested to determine conformance to heating and air conditioning performance requirements specified in 3.5.10 through 3.5.10.4.1. Heating and air conditioning tests shall be as specified in 4.3.1.1 and 4.3.1.2. Test data, including a description of all applicable and pertinent test conditions, shall be recorded and furnished to the Government representatives. If the vehicle manufacturer has previously tested an identical model vehicle with the same components, a certification and the specified test data verifying conformance to this specification may be provided in lieu of the first vehicle test.

4.3.1.1 Heating. To verify conformance to the performance requirements specified in 3.5.10.1 and 3.5.10.2, the vehicle manufacturer shall test the vehicle under actual or simulated conditions at least as stringent as specified herein. Simulated conditions must be acceptable to the Government.

4.3.1.2 Air conditioning. To verify conformance to the air conditioning performance requirements specified in 3.5.10.4 and 3.5.10.4.1, the manufacturer shall test the vehicle under actual or simulated conditions at least as stringent as specified herein. Simulated conditions shall be acceptable to the Government. Inside temperature may be calculated either with the number of specified seated passengers for the type of vehicle specified, or by the use of simulated heat, for sensible cooling. Solar radiation may be employed or simulated by calculating solar load effective in Btu, based on the square foot surface and the angle of incidence that will be exposed to the sun at the same time.

* 4.3.2 Vehicle weight. The first production vehicle shall be weighed to determine the curb weight and distribution of the curb weight on the front and rear axles. The total imposed loading on the front and rear axles shall be computed by the contractor and verified by the Government, using the curb weight, the operator weight at 165 pounds, and the payload required to provide the specified GVW. The calculated imposed loads on the front and rear axles shall be compared to the suspension, axle, and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

* 4.3.3 Road test. The first production vehicle will be examined and road (less payload) by the contractor to assure that the vehicle will operate in accordance with the contractual requirements. The road test shall be for a distance of not less than 50 miles over streets, highways and improved gravel roads. Subsequent production vehicles will be subjected to a similar road test of 10 miles duration. The water spray test (see 4.3.4) shall be conducted, on all vehicles, immediately after the road test.

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* 4.3.4 Water spray test. The first completed production vehicle shall be water spray tested after the road test. Each vehicle on the contract shall be subjected to a water spray test and shall be free of leakage after exposure to a 100 percent water spray for not less than 15 minutes. The spray shall be delivered by nozzles operating at not less than 25 pounds per square inch (psi) water pressure, sufficient in number, and so placed (approximately 4 feet from the body) to afford full coverage of sides, roof, front and rear of the body. Section by section test is acceptable. During test, doors shall be closed as in normal travel position.

4.3.5 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of the vehicle until corrective action has been taken.

4.4 Inspection of production vehicles. Each production vehicle (less payload) shall be examined and driven on approximately level road surfaces by the contractor to assure that the vehilce will operate in accordance with the contractual requirements. The contractor may authorize his sub-contractor who will perform the final assembly work on the vehicle to make such examination and tests, but this shall not relieve the contractor of his obligation to deliver an acceptable vehicle or constitute a waiver of any warranty provision. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 have been accomplished.

4.4.1 Rejection. Deficiencies of workmanship and nonconformance to any requirements of the contract shall be cause for rejection until corrective action has been taken.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

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6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use on large posts, camps, and stations; for long trips, over roads and highways encountered in the United States and overseas; for transporting adult personnel, their baggage and parcels.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and passenger capacity of vehicle required (see 1.2).
(c) Identification of appropriate service for painting and marking (see 3.1.1.1).
(d) Concealed markings, if required (see 3.1.1.1).
(e) Rustproofing, if required (see 3.1.1.8).
(f) Manual transmission, if required (see 3.4.6.2).
(g) Oil seals, if required (see 3.4.10).
* (h) Traction control, if required (see 3.4.10.1).
* (i) Disc type wheels, if required (see 3.4.11).
(j) Spare wheel or rim, if required (see 3.4.11.3).
(k) Spare tire assembly, if required (see 3.4.11.4).
(l) If passenger signal buzzer is not required (see 3.4.13.3).
(m) Individually controlled reading lights for type I, if required (see 3.4.13.4).
(n) A spotlight, if required (see 3.4.13.5).
(o) If tools are not required (see 3.4.18).
(p) If a destination sign compartment is not required (see 3.5.4.4).
* (q) If air conditioning is not required (see 3.5.10.4).
(r) Lavatory facilities, if required (see 3.5.11.1).
(s) If package racks are not required for type I (see 3.5.12).
* (t) Enclosed package racks with doors, if required (see 3.5.12.1).

6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient temperature, and in still, dry air. The factors to be used in predicting vehicle ability (see 3.3) are established as follows for the corresponding SAE J688 Truck Ability Prediction Procedure Tables:

Table 1 - Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.

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Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 2.080
Table 4	- Area Factor. 0.197
Table 5	- Velocity Factor. 432.0
Table 6	- Altitude Factor. 1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest higher 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 60 mph geared speed. For GVW and engine range beyond the range of this table, the following formula shall be used: FHP = 1 + (0.0000002 rpm) (GVW).
Table 8	- Grade Factor. 0.62
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0.

* 6.4 Subject term (key word) listing.
Air suspension
Bus, air conditioning system
Bus, motor, commercial
Bus, motor, intercity
Bus, motor, nontactical
Bus, motor, suburban
Bus, passenger seats, intercity
Bus, passenger seats, suburban
Bus, performance criteria
Torsilastic suspension

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the last previous issue.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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